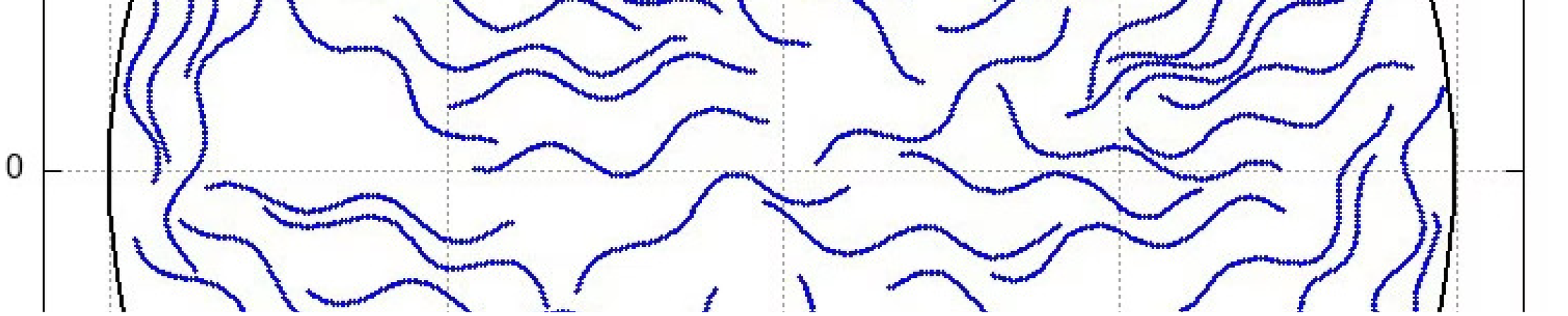

Taylor Line Swimmers in Confinement

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Course: PYD411

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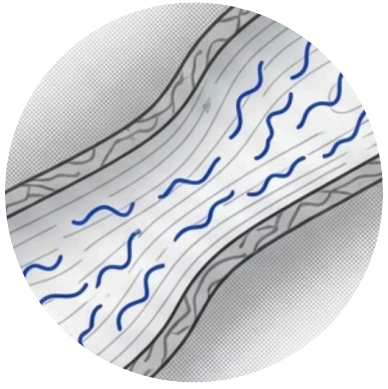
Project Aim: To study the collective dynamics of Taylor-line microswimmers confined within a circular boundary using numerical simulations.

Motivation

- **Understanding Microswimmers:** Microswimmers are self-propelled microscopic particles that continuously convert energy into directed motion and they actually interact all the time with us, so studying them is significant.



Confinement Matters

Boundaries have significant impact on collective motion and can create clustering, and boundary accumulation that are not present in bulk systems

Real-world Relevance

Real-world environments are rarely un-bounded.

Examples: bacteria in capillaries — confinement alters transport and organization.

Project Focus: This study currently ignores hydrodynamic interactions to focus on the minimal components of collective behavior: steric interactions and self-propulsion. This streamlined approach lowers computational complexity, allows for large-scale simulations, and isolates universal mechanical effects.

Computational Setup — Model

Swimmer: discrete chain of N beads with Hookean springs and bending potentials. Active propulsion via prescribed sinusoidal curvature along the filament.

Steric repulsion prevents overlap. Simulation step with Velocity–Verlet algorithm.

1 INITIALISATION

2 FORCE CALCULATION

3 STREAMING STEP

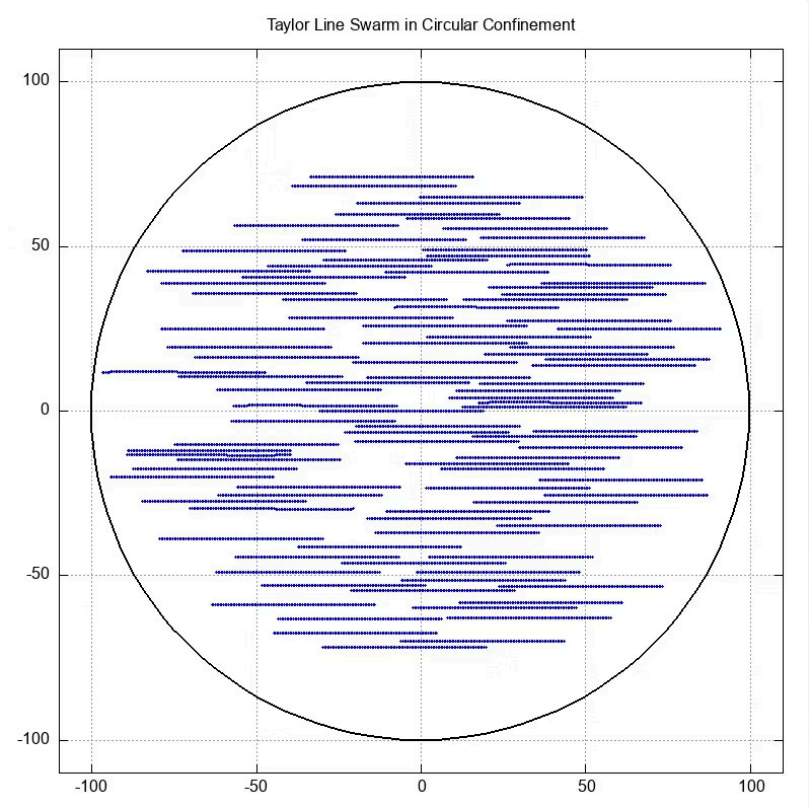
4 REPEAT STEP 2&3

Initialization of Swimmers

Swimmers are initialized at uniform random and non-overlapping positions inside a circular confinement.
If they overlaps we reassign them a new positon to begin with.

Boundary Interaction: Normal Component of bead interacting with boundary is simple reveresd.

The swimmer is assigned a constant self-propulsion velocity v0 along its orientation vector.



Force Calculation

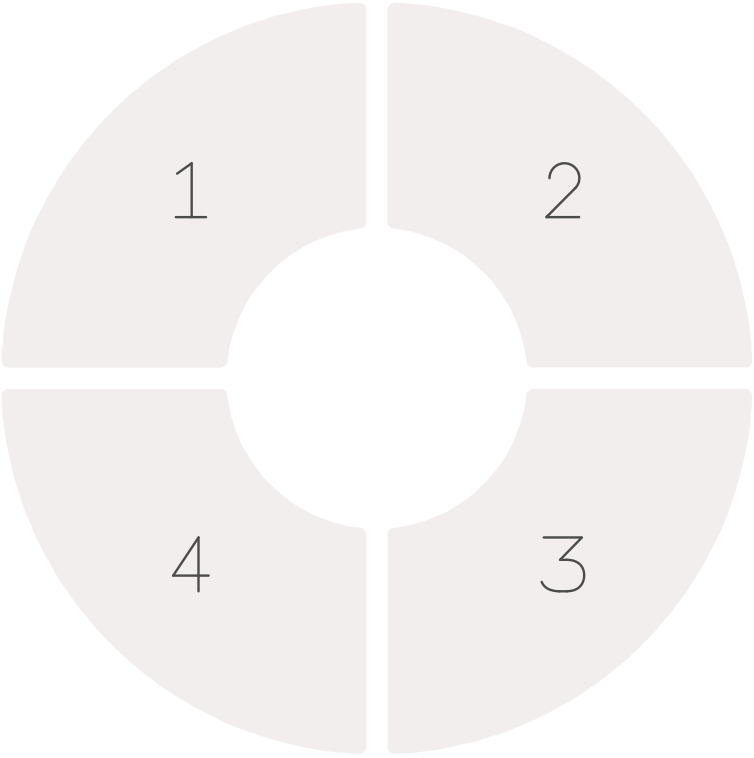
Total Force

$$\mathbf{F}_i = \mathbf{F}_i^{\text{internal}} + \mathbf{F}_i^{\text{external}}$$

Steric

$$U_{\text{steric}}(r) = 4\epsilon \left[\left(\frac{r_0}{r}\right)^{12} - \left(\frac{r_0}{r}\right)^{6} \right]$$

for $r < 2^{1/6} r_0$



Hook's Spring Potential

$$V(r) = \frac{1}{2} k \left(r - 0.5\, a_0\right)^2$$

Bending Potential

$$U_{\text{bend}} = \frac{k}{2} \sum_{i=1}^{N-1} \left|\mathbf{t}_{i+1} - R(\alpha_i)\mathbf{t}_i\right|^2$$

$$\alpha_i = l\, c(i,t),$$

$$c(i,t) = b \sin\left[2\pi\left(\nu t + \frac{2i}{N}\right) + \phi\right]$$

Streaming Step

Velocity–Verlet algorithm updates positions and velocities each timestep:

1

Position

$$\mathbf{r}_i(t+\Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\Delta t + \frac{\mathbf{F}_i(t)}{2m}\Delta t^2$$

—[2]

2

Velocity

$$\mathbf{v}_i(t+\Delta t) = \mathbf{v}_i(t) + \frac{\mathbf{F}_i(t) + \mathbf{F}_i(t+\Delta t)}{2m}\Delta t$$

—[2]

After updating the position, forces of all swimmers are recalculated and the same is repeated.

Results — Simulation Snapshots

Simulations show swimmers concentrate near boundaries. Increasing swimmer number enhances crowding and boundary accumulation (Figures 1–2).

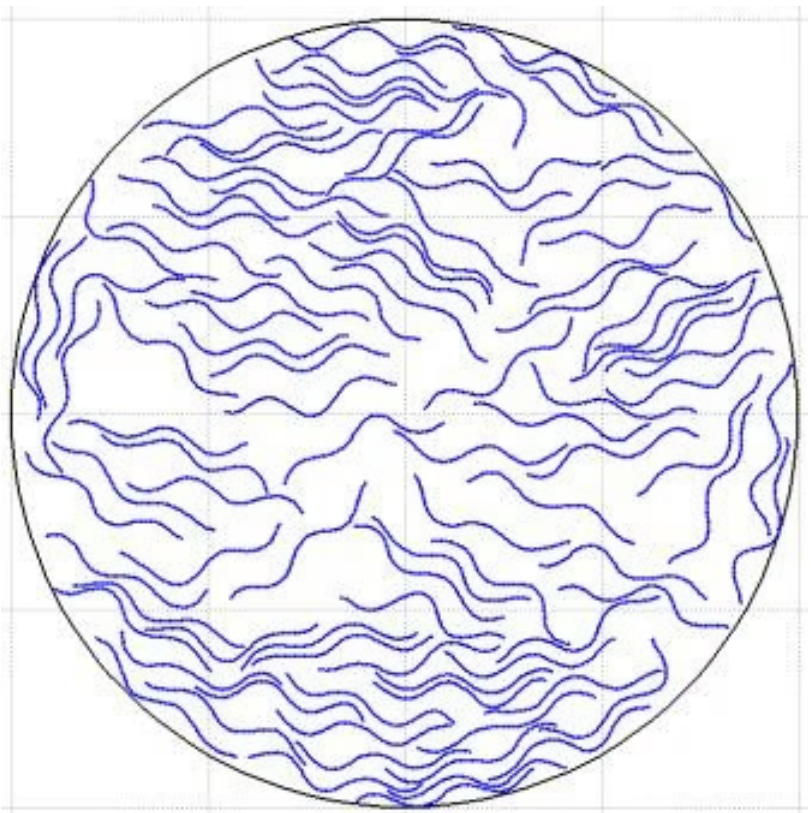


Figure 1 (Radius of circular confinement is 100a0)

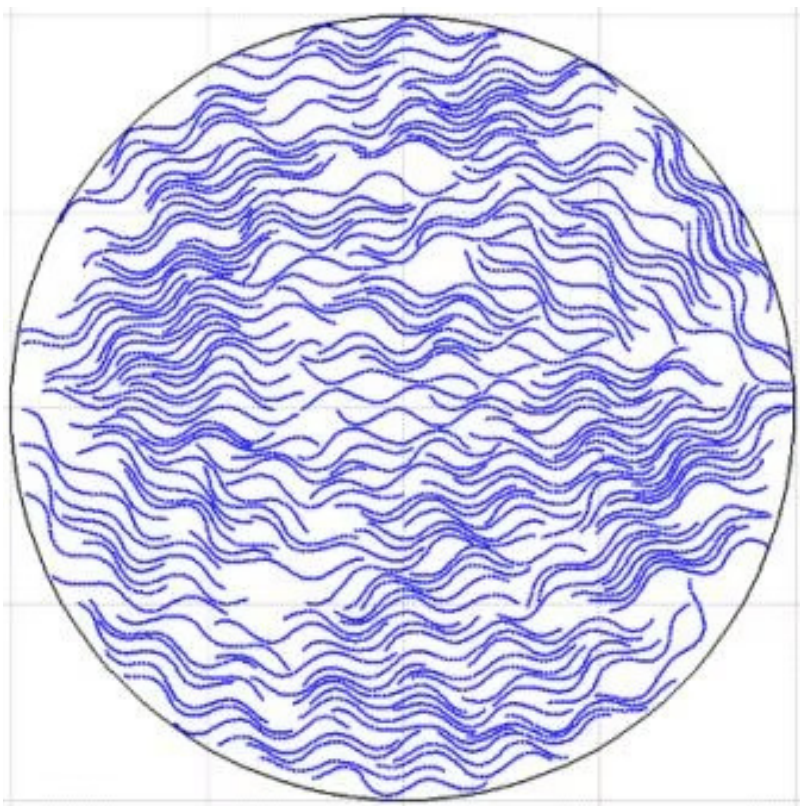
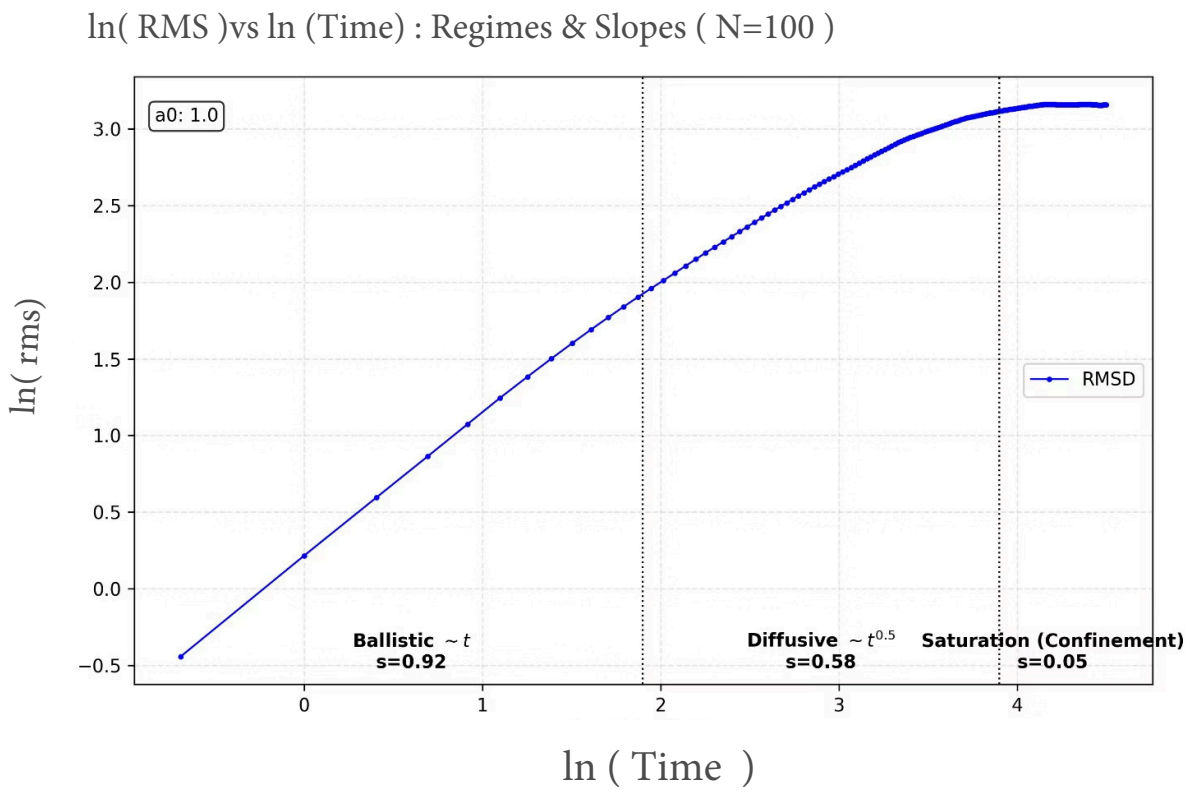


Figure 2 (Radius of circular confinement is 100a0)

Results

RMS displacement of swimmers shows three regimes: Ballistic → Sub-diffusive → Saturation .



- **N=100, Beating freq = 0.001 Hz**

  Ballistic slope ≈0.9 then ≈0.58 for Diffusive then ≈0.05 as saturing

- **Swimmers intially moved in a st line to reach boundary, then try to diffuse and then due to confinement, saturation is observed.**

Results

Velocity Correlations vs ln(Time(s)) Curves

Velocity autocorrelation decays exponentially toward zero on log time scales, indicating loss of memory of system. Higher N and frequency alter smoothness and oscillatory noise (Figures 4–7).

N=100, Beating Freq. = 0.001 Hz

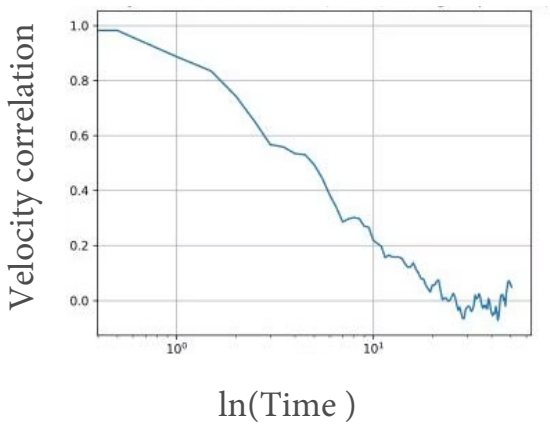


Figure 4

N=100, Beating Freq. = 0.005 Hz

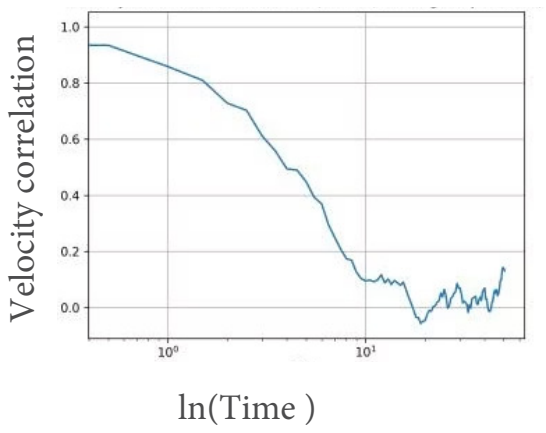


Figure 5

N=200, Beating Freq. = 0.001 Hz

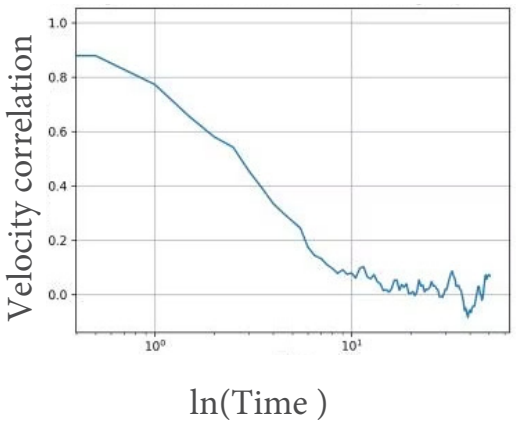


Figure 6

N=200, Beating Freq. = 0.005 Hz

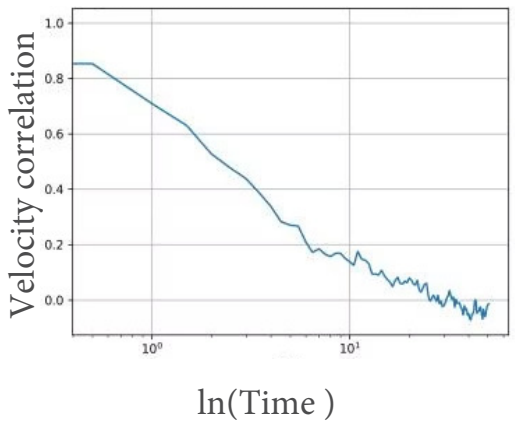


Figure 7

Future Scope, Acknowledgements & References

Future Work

Include thermostat, vary beating frequency and might introduce thermal fluctuations, and perform density scans to build a phase diagram of confined active matter.
Calculate Segarigation Index to study segeregation dynamics.
Might Introduce hydrodynamic interactions.

Acknowledgements

We thank Prof. Sujin B. Babu for guidance and valuable discussions throughout this project.

References

[1] Adyant Agrawal* and Sujin B. Babu ‡ Self-organization in a bimotility mixture of model microswimmers (2018).

[2] Journal of Applied Mathematics and Physics, Vol.7 No.9, September 28, 2019